A High Spatial Resolution Aerial Image Dataset and an Efficient Scene Classification Model

Yi Lin, Lin He, Daiqi Zhong, Yufei Song, Lujia Wei, and Liang Xin

***Abstract*— In the past decade, deep neural networks have been widely introduced into remote sensing image interpretation tasks. Exploring how to improve the model at some critical points, such as making it more suitable for remote sensing data, is still lacking, which makes more sense than developing stacked and transformed network structures, as is the expertise of scholars in the field of deep learning. To tackle this issue, this article proposes a fixed-length sequence-vision reformer (FLS-ViR) model. Relying on an advanced attention encoder with a reversible residual network structure, the presented model drastically decreases the spatial complexity. In contrast to most models that take input image sizes with 200+ pixels as edges, it will help us allow the input of large-size images under limited equipment conditions, thereby making fuller use of high spatial resolution data information. The introduction of a self-supervised method further improves the detection accuracy. Meanwhile, we built a high spatial resolution aerial image dataset for scene classification, which is superior in terms of high interclass similarity and diversity of image variations. To validate and evaluate the method and dataset, we performed a comparison of accuracy and memory consumption on a similar aerial dataset and attention-base models. The effectiveness of our model and its fitness on remote sensing datasets with a limited sample size is proved. Meanwhile, the investigation of the heat map demonstrates that increasing the size of the input image has a positive effect on optimizing the features captured by the model for remote sensing image scenes.**

***Index Terms*— Aerial image dataset, scene classification, space complexity, transformer.**

I. Introduction

CLASSIFICATION is a foundational and essential task in remote sensing image interpretation, which is meant to implement meaningful semantic annotation of images. At present, the classification of remote sensing images is mainly performed at pixel level [1] and object level [2], both of which complete the modeling of remote sensing scenes in a bottom-up manner and identify features by classifiers [3]. However, these classifications lack thematic meaning [4], and it is difficult to reveal useful semantic information in images similar to remote sensing scenes [5], which leads to the divergence between low-level data and high-level semantics [6], i.e., the "semantic gap." Scene-level classification, on the other hand, can understand and analyze the deep semantic information of the number and spatial relationship of objects in the image consisting of multiple classes of features [7].

The concept of a "scene" on a remote sensing image usually refers to an image area that reflects explicit semantic information about the type of land surface features [8]. Remote sensing image scene classification is the annotation of semantic categories for remote sensing images covering multiple land cover types or ground objects [9]. It has been widely utilized in tasks such as feature detection, object tracking, and image interpretation [10] as well as urban planning [11], disaster monitoring [12], and other fields.

Depending on the features used for classification, traditional scene classification methods can be grouped into two categories: methods based on low-level features and methods based on mid-level semantic features [13]. Scene classification based on low-level features applies features such as area, spectrum, and texture of objects in the scene. In this approach, specific scenes in a specific study area are usually analyzed in more detail, but the generalizability of the method is low and the detection accuracy needs to be improved. Classification methods based on mid-level semantic features include Bag of Visual Vocabulary (BoVW) [14], spatial pyramid matching (SPM) [15] and other feature encoding methods [16], [17]. Such methods combine independent base features to derive more advanced feature representations, but the lack of partial information causes some limitations of such methods [18].

Given the features of deep convolutional neural networks (CNNs) that can automatically learn high-level semantic information from annotated samples through an end-to-end supervised learning process and the compelling results achieved in the field of computing. Methods for remote sensing image scene classification based on high-level features have been gradually developed in recent years. Typical deep neural network models including deep belief networks (DBNs), stacked autoencoder networks (SAEs) and CNNs [19] have been widely introduced in remote sensing image scene classification tasks. An effective feature selection algorithm can improve the final results of scene classification. Compared with traditional feature-based algorithms, deep neural network models driven by data can model the high-level semantic content of remote sensing images and perform better in scene classification tasks [20]. Therefore, scholars in remote sensing

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continue to pay attention to the latest advances in the field of deep learning [21].

Due to the complexity and diversity of features presented, a large amount of redundant information is presented in remote sensing images. Usually, multiple land covers share the same scene, and various scenes contain the same land covers. These make it necessary to comprehend features over a broad scope of remote sensing images [22]. While the CNN-based methods which are widely used to solve vision problems do not have sufficient differentiation for redundant and critical information and usually can only utilize local connections for extraction while ignoring the long-distance dependence within the scene [23]. Therefore, another competitive neural network based on Attention is gradually being widely used for remote sensing image scene classification tasks [24] and Transformer is one of them.

In recent years Transformer networks have excellent performance in language tasks [25], and have also been introduced in vision tasks, such as image quality evaluation, image instance segmentation [26] and video target detection tasks [27]. In 2020, Dosovitskiy et al. [28] proposed the vision transformer (ViT) model, which applies the Transformer to the field of image scene classification. ViT separates the input image into slices, and uses the linear embedding sequence of the slices as the input to the Transformer encoder, preserving the same network processing of the image slice as natural language processing (NLP) does for words. ViT has achieved better detection results on public datasets for several image classification tasks, and has soon extended and widely used in computer vision fields such as scene classification after its proposal [21], [29], [30]. There are many scholars who applied ViT to experiments on remote sensing image scene classification tasks [31], such as the article [21] on the experimental evaluation of ViT models for remote sensing image scene classification and the fusion algorithm [32] of CNN and ViT for the advantages of both algorithms in information representation.

With the development of the Transformer-based model, more multimodal models [33] have been generated from the foundation model [34], and these models complement information from different modalities and achieve promising results on downstream tasks [35]. The effectiveness of multimodal models has also been demonstrated in remote sensing image processing [36], [37]. These state-of-the-art models focus on the application of multimodal data on the optimization of feature extraction and the upgrade of accuracy, which can be understood as an improvement of the data-driven models with the same goal as some of the model-driven approaches adjusting the structure of the network [38]. Different from concentrating on how to increase accuracy, in this article we would like to exploit the contradiction between the Transformer-based method and the remote sensing data, and then attempt to solve it.

As scholars have pointed out, Transformer has some limitations, including the fact that the spatial complexity grows quadratically with the input sequence length, leading to memory bottlenecks when modeling fine-grained long-time sequences, and also making the computation less efficient. For remote sensing images with high resolution usually require more pixels to describe a meaningful scene. Many scholars reduced the amount of input data by image downsampling and thus controlling the input length to alleviate the problem of computer resource constraints. However, this causes a certain degree of information loss and overshadows the data strength of high-resolution images covering rich feature detail. Several papers have proposed effective methods for mitigating the Transformer in terms of spatial complexity growth, such as the LogSparse Transformer model [39], which only needs to calculate $\log L$ times dot product per cell in each layer and only needs to be superimposed to the $\log L$ layer, and the model can access the information of each cell, thus reducing the space complexity to $O(L(\log L)^2)$. In contrast, the Reformer model [40] reduces the space complexity to $O(L \log L)$ by an approximate attention calculation based on a locally sensitive hashing algorithm.

Meanwhile, the ViT model for image processing, like Transformer for NLP, suffers from the appetite for data. Some researchers have attempted to address this problem by data augmentation [41] to expand data based on limited samples. While the acquisition and labeling of remote sensing data are more difficult than natural images, the solution to the data demand of the model cannot be carried out from the data augmentation aspect alone. Self-supervised deep neural network training models [42] have been consistently proposed in recent times, and studies have shown that self-supervised networks can attain satisfactory high-level semantic features of images [43]. Even if it is not possible for most scholars to apply a foundation model [44] for training and thus obtain a highly robust feature representation, shallow self-supervised models can be effective for the accuracy improvement of vision tasks on datasets with a small number of samples [45].

To address the above problems, a scene classification model for high-resolution aerial photogrammetry data is constructed. First, a reversible residual structure is introduced to reduce the memory occupation, and then a self-attention constrained by the length of the sequence is proposed to reduce the spatial complexity of the algorithm. It will enable the model to be trained under limited equipment conditions, while large-size input images could be allowed, which will enhance the utilization of the high-resolution data. The pretraining weights of the model are obtained through a self-supervised pretraining method, which is employed as an auxiliary strategy to improve the accuracy of our model training on the samples. We also provide a publicly available high-resolution aerial image dataset containing 27 scene categories. The dataset contains four bands with a spatial resolution of 0.1 m, as well as high interclass similarity and diversity of image variations.

The article will be organized as follows: the provided dataset is introduced in Section II. The proposed model is detailly described in Sections III-A and III-B. The evaluation metrics are expressed in Section IV-A. Section IV-B gives the testing and evaluation of the proposed dataset. Performances of our novel model are presented in Section IV-C, including the parameter effects, the comparison and the analysis of different scene classes.

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TABLE I
REMOTE SENSING DATASETS FOR SCENE CLASSIFICATION

Dataset	Number of categories	Number of Images	Spatial Resolution/m	Image Size	Number of Bands	Data Source	Year of publication	Production Team
NASC-TG2	10	20000	100	128x128	17	Tiangong-2	2021	CAS
BIGEARTHNET	43	590326	10/20/60	120x120/60x60/20x20	13	Sentinel-2	2018	TUB
SIRI-WHU	12	2400	2	200x200	3	Google Earth	2016	WHU
WHU-RS19	19	1005	0.5	600x600	3	Google Earth	2011	WHU
AID	30	10000	0.5-8	600x600	3	Google Earth	2017	WHU, HUST
NWPU-RESISC45	45	31500	0.2-30	256x256	3	Google Earth	2017	NPU
UC MERCED	21	2100	0.3	256x256	3	USGS	2010	UC
RSI-CB	35/45	24000+/36000+	0.22-3	256x256/128x128	3	Google Earth, Bing Maps	2020	CSU
RSC11	11	1232	0.2	521x512	3	Google Earth	2015	CAS

II. DATASET

In order to build a deep neural network model applicable to remote sensing image scene classification tasks, scholars have focused on Transfer Learning by applying the computer vision dataset [46], which has a large scale of image data volume and rich scene variation. Considering the differences in features between natural images and remote sensing images, many research teams have proposed a variety of remote sensing image datasets in order to apply remote sensing images for adequate training of deep networks. However, the existing remote sensing datasets have certain limitations, such as few scene types, small number of images, few spectral numbers, low spatial resolution, and lack of scene variation and diversity. Therefore it is necessary to extend and enrich the remote sensing image scene classification datasets.

Scholars have made a lot of efforts on extending the volume and diversity of remote sensing image datasets. The datasets proposed in the early years [47], [48] mostly use images from Google Earth as the data source. The single data source leads to great similarity and redundancy between different datasets, mainly in the fact that these datasets have a single spatial scale. Datasets proposed in recent years are more abundant, the remote sensing images produced by Sentinel-2 [49] and Tiangong-2 [4] as data sources enrich the types and features of remote sensing scene classification datasets. With the remote sensing data being enriched, high spatial resolution becomes a further requirement.

The detailed information obtained by researchers from remote sensing image datasets is closely related to the spatial resolution of the datasets. The high spatial resolution makes the semantic information of the image clearer and therefore more conducive to scene recognition. In the past few years, many research teams have proposed distinctive remote sensing image datasets for scene classification. In this section, representative datasets are reviewed in order of spatial resolution and summarized in Table I.

The 10–100-m spatial resolution dataset contains the NASC-TG2 dataset [4] with multispectral imagery at 100-m spatial resolution, a large amount of data, and large intraclass variability and high interclass similarity. The BIGEARTHNET dataset [49], the Technical University of Berlin released in 2018, as a Sentinel-2 dataset with spatial resolutions of 10, 20, and 60 m, contains images from ten European countries. There are also datasets with meter-level spatial resolution, such as the SIRI-WHU dataset [7] released by the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing of Wuhan University in 2016, which has a spatial resolution of 2 m and mainly covers urban areas in China. Remote sensing image datasets that reach the decimeter-level spatial resolution, like the WHU-RS19 dataset [47] released by the same Laboratory of Wuhan University in 2011, whose spatial resolution can reach 0.5 m. The AID [50] has a spatial resolution ranging from 0.5 to 8 m and an image size of 600×600 pixels. The UC MERCED dataset [51], which has a spatial resolution of up to 0.3 m, contains highly overlapping and uses classes with different structural densities at the same time. As a classical common classification dataset, the UC MERCED dataset is often used for algorithm validation and model pretraining applications. NWPU-RESISC45 [52], with spatial resolutions ranging from 0.2 to 30 m, is richly varied in terms of lighting, background, shading, etc. RSI-CB dataset [53], with the spatial resolution of the dataset images mainly concentrated between 0.22 and 3 m, constructs a global-scale benchmark for large-scale remote sensing image classification. It contains sub-datasets with two image sizes of 256×256 and 128×128. In 2015, CAS released the RSC11 dataset [48], a high-resolution remote sensing images dataset with a spatial resolution of 0.2 m. Some scene classes in the dataset are visually very similar, which increases the difficulty of identifying scene images.

Although the size of remote sensing image datasets is getting larger, it is still difficult to match the size of computer vision datasets. This is due to the fact that remote sensing images are much more difficult to acquire than ordinary images, and the complexity of implementing semantic annotation on remote sensing images is also much more complicated than natural images. Therefore, the size of remote sensing image datasets hardly surpasses that of computer vision datasets. Some researchers have made thematic datasets for specific remote sensing scenarios, such as the dataset NaSC-TG2 [4] for natural landscapes. This provides scholars with the idea that if remote sensing datasets cannot match computer vision datasets in terms of scale, it is better to consider the practical application value.

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TABLE II
LIST OF CATEGORIES IN TJAID

Class No.	Type of ground feature	Class No.	Type of ground feature	Class No.	Type of ground feature
1	Industrial land	10	Artificially planted wood	19	Railway
2	Storage land	11	Forest	20	Bridge
3	Agricultural greenhouse	12	Commercial land	21	Parking
4	Cropland	13	High-rise buildings	22	Airport
5	Uncultivated field	14	Low-rise buildings	23	Boat
6	Bare land	15	Villa zone	24	Golf course
7	Mudflat	16	Rural residential land	25	Playground
8	River	17	Intersection	26	Basketball court
9	Pond	18	Road	27	Park



Fig. 1. Samples of 27-category scene images. (From left to right) Each column of images corresponds to an image index number of 1–27 in Table II.

We established a competitive dataset, named Tongji Aerial Image Dataset (TJAID), which is an aerial remote sensing image dataset with high spatial resolution and rich semantics. These images come from the aerial images of about 14.6 km^2 of Shanghai city by the Shanghai Institute of Surveying and Mapping in 2021. The dataset is oriented to scene classification applications, containing 1387 images with 27 scene categories, with a single image size of 1024×1024, and a spatial resolution of 0.1 m, including blue, green, red, and near-infrared bands. All image categories are manually labeled, and Fig. 1 shows some images of the 27 scene categories of the TJAID. The number of sample images for different aerial scene types (see Table II) is small. Compared with existing scene classification datasets, TJAID has three significant features as follows.

A. High Interclass Similarity and High Intraclass Variability

The poor diversity and high homogeneity of image content in some datasets make the deep neural network approach easily reach saturation classification accuracy. For this reason, we added some scene classes with high interclass similarity, e.g., artificially planted woods and forest (columns 10 and 11 in Fig. 1). We also added images with large differences in the same scene category, for example, two industrial lands containing factories and storage tanks, respectively (column 1 in Fig. 1). Different scenes have high semantic overlap and the same scene has semantic differences, which increases the difficulty of reaching saturated classification accuracy and challenges the optimal accuracy that can be obtained by the deep neural network models.

B. Diverse Image Variations

The large number of category types and intraclass variability within the same category give the dataset rich image variation. In aerial imagery, objects in the same class of scenes may be presented at different scales and angles due to the dramatic effect of different imaging conditions (e.g., altitude and flight pitch angle) on the appearance of objects imaged in the class of scenes, especially artificial buildings or landscapes with a certain height. Images in the TJAID are diverse in variation in panning, viewpoint, pose and appearance, illumination, occlusion, etc.

C. High Spatial Resolution and Four Spectral Bands

Compared to other datasets, TJAID is rarely compatible with high spatial resolution and over three spectral bands, which means the images in the dataset can reflect more detailed spatial information of the features and richer spectral information. With the high spatial resolution, the geometric structure of clear scenes in aerial images poses additional challenges to the classification task. The 0.1-m spatial resolution allows the scene images to clearly present details about ground features such as water bodies, vegetation, roads, buildings and vehicles. At the same time, the addition of the near-infrared band helps to provide more characteristics for scene understanding and identification, and therefore clear and rich semantic information of the scene can be obtained.

III. METHODOLOGY

This article proposes a fixed-length sequence-vision reformer (FLS-ViR) model (in Fig. 2) for the high-resolution

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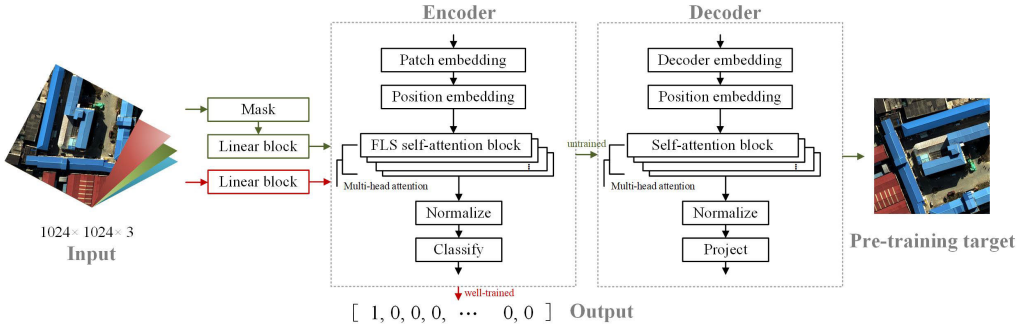


Fig. 2. FLS-ViR model scene classification flowchart. Pretraining is achieved through a self-supervised method and scene classification is carried out through an encoder with a classifier.

aerial image classification task. The pretraining network employs a self-supervised method to construct a multi-head self-attention encoder with fixed length sequences, followed by an implementation of a decoder block containing a ViT model. The well-trained encoder is used to achieve scene category recognition by a classifier.

A. Vision Reformer

An image-codable Reformer model based on self-attention will be realized which is the basis of our model. First of all, an image-to-sequence conversion method in ViT [28] is introduced into the Reformer model. An input image is cut into patches and linearly expanded into sequences. After accessing an encoder and classifier, a vision reformer (ViR) model can be implemented for scene recognition tasks.

The Transformer is a simple and efficient feedforward neural network model based on attention, consisting of alternating layers of MSA and multilayer perceptron (MLP) blocks [54]. The MSA uses different learnable linear projections to project sequence values onto their respective corresponding dimensions h times (h is the number of heads). The self-attention is executed in parallel on each linear projection and can be expressed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where softmax is the softmax logistic regression model, Q, K and V are the corresponding three input sequences, and d_k is the dimension corresponding to the K sequence.

The Reformer model is an improved network proposed for solving the problem of large memory consumption of the Transformer model, and its main scheme is to apply locality sensitive hashing (LSH) and reversible residual layers.

Setting the sequence length of the model input to L, the space complexity of the Transformer model based on the self-attention grows quadratically with L, while the Reformer model reduces it to $O(L \log L)$ by combining the LSH method. Neglecting the scale parameter $(d_k)^{1/2}$, the Attention(Q, K, V) at the single query position i can be expressed as o_i

$$o_i = \sum_{j \in P_i} \exp\left(q_i \cdot k_j - z(i, P_i)\right) v_j \tag{2}$$

where z denotes the partition function (i.e., the normalization term in softmax), P_i denotes the set that the query at position i attends to, and h denotes the hash bucket. Unlike the Transformer model where $P_i = j : i \geq j$ and attention is performed in all vectors, the Reformer model introduces the LSH method, which can help to find nearest neighbors quickly in high-dimensional spaces, into the self-attention by only allowing attention within a single hash bucket, i.e., $P_i = \{j : h(q_i) = h(k_j)\}$. The residual network is applied in Transformer for solving the gradient disappearance problem, but it needs to store the activation of each layer in memory to calculate the gradient during backpropagation, thus increasing the memory consumption of the algorithm. Set the input data for the encoder structure to x_1, and the encoder block in Transformer will be

$$\begin{aligned} y_1 &= x_1 + \text{attention}(x_1) \\ y_2 &= y_1 + \text{feedforward}(y_1) \end{aligned} \tag{3}$$

where y_1 is the output of the attention layer attention (containing a normalization layer and a self-attention layer) and y_2 is the output of the feedforward layer feedforward (containing a normalization layer and an MLP layer). In contrast, Reformer applies a reversible residual block instead of a residual block, which allows the algorithm to back-propagate without storing the activations in memory, facilitating more efficient memory utilization. In addition, an encoder layer in Transformer produces an output for a single input, while a reversible layer works on pairs of inputs (x_1, x_2) and corresponds to a pair of outputs (y_1, y_2).

The image $x \in R_{H \times W \times C}$ with the size of (H, W) and the channel of C, is cut into patches $x^p \in R_{(P \times P \times C) \times N}$ with the size of (P, P), where the number of patches is $N = HW/P^2$. The patches are linearly projected by the patch embedding projection E into D dimensions, as the sequence z. A class token and a learnable position embedding E_{pos} are also added into the sequence z. The reversible residual layers in the Reformer encoder consist of L layers of multihead LSH self-attention and MLP blocks with chunk operation CHUNK and the inputs are expressed as z^1 and z^2. The reversible residual connection is applied after every block. In the end, the scene category label y of input image x will be obtained by the classifier CL, after performing a Layernorm operation LN. ViR for image classification can be expressed as

$$z_1 = \left[x_{\text{class}}; E\left(x_p^1\right); E\left(x_p^2\right); \ldots; E\left(x_p^N\right)\right] + E_{\text{pos}}$$
$$E \in R_{(P \times P \times C) \times D}, E_{\text{pos}} \in R_{(N+1) \times D}$$

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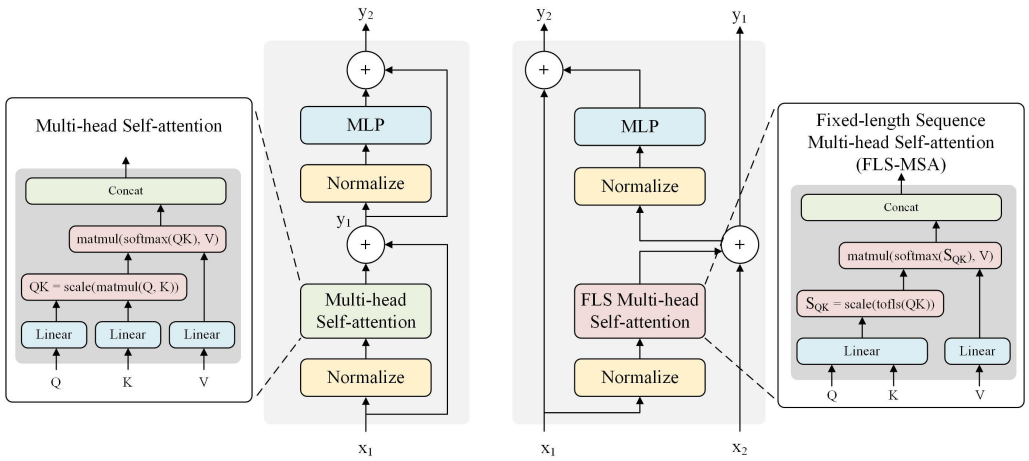


Fig. 3. Comparison of (a) ViT encoder block and (b) FLS-ViR encoder block.

$$
\begin{aligned}
z_l' &= \mathrm{MSA}\left(\mathrm{LSH}\left(z_{l-1}^2\right)\right) + z_{l-1}^1, \quad l = 2, 3, \ldots, L \\
z_l &= \mathrm{CHUNK}\left(\mathrm{MLP}\left(z_l'\right)\right) + z_{l-1}^2, \quad l = 2, 3, \ldots, L \\
y &= \mathrm{CL}(\mathrm{LN}(z_L)).
\end{aligned} \tag{4}
$$

B. FLS-MSA Encoder

Considering the space complexity reduction of the algorithm with trainable modules instead of traditional methods, in our approach, the fixed-length sequence-MSA (FLS-MSA) encoder block will replace the MSA encoder block, which consists of an attention layer and a FeedForward layer. The attention layer contains a normalization layer and an FLS-MSA layer, and the FeedForward layer contains a normalization layer and an MLP. The FLS-MSA encoder blocks are connected in the form of the reversible residual network [55]. The activation of each encoder layer can be reconstructed from that of the next encoder layer. For a pair of inputs (x_1, x_2) and outputs (y_1, y_2), an $\mathrm{Attention}_{\mathrm{fls}}$ layer $\mathrm{attention}_{\mathrm{fls}}$ contains a normalization layer and a fixed-length sequence self-attention layer, and the feedforward layer has the same definition as feedforward in (3)

$$
\begin{aligned}
y_1 &= x_1 + \mathrm{attention}_{\mathrm{fls}}(x_1) \\
y_2 &= x_2 + \mathrm{feedforward}(y_1).
\end{aligned} \tag{5}
$$

In FLS-ViR model, MSA, which contains the LSH operation as a machine learning method, will be removed and replaced by FLS-MSA, a neural network solution. That is, z_l' will be replaced by $\mathrm{FLS\text{-}MSA}(z_{l-1}^2) + z_{l-1}^1$ in (4).

Transformer uses three different linear layers projecting the input data into Q, K and V with different learnable parameters, while experiments on the self-attention show that using the same parameters for Q and K does not affect the performance of the Transformer [40], so sharing parameters are also used in the proposed approach. The sequence will be mapped to a fixed-length sequence by a trainable linear projection (tofls in Fig. 3). With the approach of introducing fixed-length sequences, the sequences can be selected by network structure without the help of complex data retrieval methods like LSH, and the space complexity can be reduced to $O(1)$. Attention will be carried out between the fixed-length sequence S_{QK} and V

$$
\mathrm{FLS\text{-}Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{S_{QK}}{\sqrt{d_k}}\right)V. \tag{6}
$$

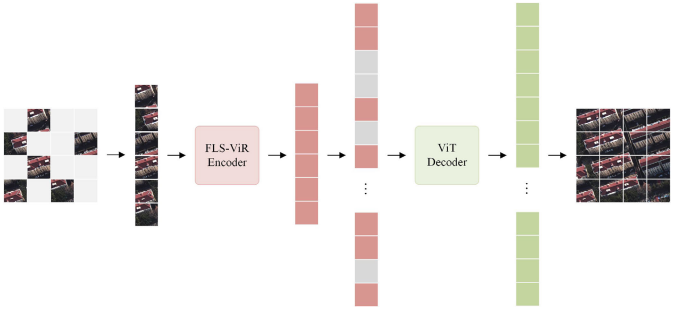


Fig. 4. Flowchart of pretraining based on the self-supervised method.

C. Pretraining Based on Self-Supervised Method

In our experiments, we found that satisfactory training accuracy cannot be achieved based on pretrained models or randomly initialized parameters, so the pretraining process of the proposed model adopts an efficient self-supervised training framework, masked autoencoders (MAEs) [45] for improving training accuracy. As shown in Fig. 4, the image data is first sliced into patches and the information is removed by random sampling of some of the patches. The FLS-MSA encoder only processes the subset of data, i.e., the remaining input data after being masked. The encoder embeds patches by trainable linear projections with added class tokens, patch embeddings and positional embeddings. The encoded visible patches and learnable mask tokens are fed into the decoder together. The main component of the decoder is a shallow ViT neural network, which outputs a vector with the same number of pixels as the input data, and the vector is finally projected as an image by linear expansion. The mean-square error of the reconstructed image and the original image in pixel space will be calculated by a loss function and back-propagation is performed to update the parameters. The decoder will be discarded after pretraining. The parameters of the encoder

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obtained by pretraining will be entered into the FLS-ViR model as initialization weights for the training process.

IV. Results and Discussion

A. Evaluation Metrics

In this article, three accuracy evaluation metrics, TOP-1, TOP-3 and TOP-5 error rates [56], are selected to evaluate the scene classification results. The probabilities of n ($n = 27$ for TJAID) categories produced by the classifier are ranked from highest to lowest, defined as $[pb_1, pb_2, \ldots, pb_n]$, the corresponding category label is $[cb_1, cb_2, \ldots, cb_n]$. TOP-1 means the number of images, where the category label cb_1 corresponding to the highest probability matches the correct label, as a percentage of the total number of images. TOP-3 refers to the number of images, where one of the category labels (cb_1 *or* cb_2 *or* cb_3) corresponding to the top three probabilities is the same as the correct label, as a percentage of the total number of images. TOP-5 is the number of images, where one of the category labels (cb_1 *or* cb_2 *or* cb_3 *or* cb_4 *or* cb_5) corresponding to the top five probabilities is the same as the correct label, as a percentage of the total number of images

$$\begin{aligned}
\text{TOP-1} &= \left(\text{bool}\left(cb_1^1 = rb^1\right) + \text{bool}\left(cb_1^2 = rb^2\right)\right. \\
&\quad \left. + \ldots + \text{bool}\left(cb_1^i = rb^i\right)\right)/M \\
\text{TOP-3} &= \left(\text{bool}\left(cb_1^1 = rb^1 || cb_2^1 = rb^1 || cb_3^1 = rb^1\right)\right. \\
&\quad + \text{bool}\left(cb_1^2 = rb^2 || cb_2^2 = rb^2 || cb_3^2 = rb^2\right) \\
&\quad \left. + \ldots + \text{bool}\left(cb_1^i = rb^i || cb_2^i = rb^i || cb_3^i = rb^i\right)\right)/M \\
\text{TOP-5} &= \left(\text{bool}\left(cb_1^1 = rb^1 || cb_2^1 = rb^1 || cb_3^1 = rb^1\right.\right. \\
&\quad \left. \times || cb_4^1 = rb^1 || cb5^1 = rb^1\right) \\
&\quad + \text{bool}\left(cb_1^2 = rb^2 || cb_2^2 = rb^2 || cb_3^2 = rb^2\right. \\
&\quad \left. \times || cb_4^2 = rb^2 || cb_5^2 = rb^2\right) \\
&\quad + \ldots + \text{bool}\left(cb_1^i = rb^i || cb_2^i = rb^i || cb_3^i = rb^i\right. \\
&\quad \left.\left. \times || cb_4^i = rb^i || cb_5^i = rb^i\right)\right)/M
\end{aligned} \tag{7}$$

where, M represents the number of images in the dataset and $i = 1, 2, .., M$. bool is a Boolean operation that takes the value of 1 if the equation holds, and 0 if it does not. || indicates the OR operation.

B. Performance of TJAID

In order to verify the usability of the proposed dataset, we conducted a comparative algorithm performance test on our dataset and the AID [50], which is also an aerial image dataset and has an approximate spatial resolution and a similar number of categories.

Since the total amount of the AID is much higher than our dataset, we further selected scene categories from the AID with the same number of categories as our dataset, and evenly distributed the same number of images into each category as the total amount of our dataset. The category types and the images in each category were randomly selected. This sub-dataset of AID is called AID_MINI in the article. We selected two representative CNN-based models, ResNet [57] with 50 layers and DenseNet [58] with 169 layers, as well as ViT to train and test for scene classification on AID, AID_MINI and our dataset. The training, validation and test sets have a data volume ratio of about 6:3:1. The initialization weights used in the training are the training parameters obtained by training the model on the ImageNet-21K dataset. The obtained training and testing results are displayed in Table III. The highest validation accuracy of the model in training is marked as MaxVal, the accuracy obtained on the test set is marked as Test,

The training and testing accuracies for the AID were higher than 90% for both CNN-based models and the ViT model, while the results for the DenseNet and ResNet models on the AID_MINI dataset were both around 60%, and the training and testing accuracies for the ViT model were 80% approximately. For the TJAID, the testing and training accuracies of the DenseNet model were slightly above 70%, while the ResNet model was close to 90%, and the experimental results obtained by the ViT and ResNet models were similar.

It can be seen that the relative differences in experimental accuracy presented on the AID and TJAID are similar for the three network models. Accordingly, both the TJAID and AID are effective for the training and testing of neural network models. And the experimental results on the AID are all higher than that on the TJAID which indicates that in contrast to AID, TJAID is more challenging for models.

For the AID_MINI dataset with the same amount of data as the TJAID, the difference in experimental results between the ResNet and ViT models is larger than the other two groups of experiments. The training accuracy of the ViT model is 15% higher than that of the ResNet model, and the test accuracy of the ResNet model is more than 20% lower than that of the ViT model. For the other two groups of experiments, the difference in experimental accuracy between these two models is around 2%. Combining the lower model training and testing accuracy, compared with TJAID, with the same amount of data, AID could not obtain the same effective experimental results as TJAID for deep neural network models to achieve an accurate assessment of models' pros and cons.

Based on the above comparison experiments and analysis with the AID and AID_MINI datasets, our proposed aerial image dataset is both effective and challenging for the deep network model in scene classification, which relies on diverse image variations in the dataset, as well as its high interclass similarity and high intraclass variability.

C. Scene Classification Results and Discussion

To investigate our proposed scene classification method comprehensively, we implemented three accuracy metrics to assess the scene classification results on 27 categories of the TJAID. ViT and ViR models were taken as comparison algorithms to show the superiority of the FLS-ViR model in terms of reduced memory usage and improved accuracy, respectively. The classification effects presented on different scene categories are also compared and analyzed.

1) Nontrainable Sequence Length Effects: In the fixed-length sequence attention block, the input sequence QK is projected as a fixed-length sequence, and one of the main parameters in our model is the length of the fixed-length sequence, which affects the amount of effective information

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TABLE III
MAXIMUM VALIDATION ACCURACIES AND TESTING ACCURACIES OF RESNET, DENSENET, AND VIT MODELS ON AID, AID_MINI, AND TJAID

	AID		AID_MINI		TJAID	
	MaxVal/%	Test/%	MaxVal/%	Test/%	MaxVal/%	Test/%
ResNet_50	92.07	92.50	64.75	61.19	87.20	89.63
DenseNet_169	90.77	91.40	60.54	58.90	73.87	71.85
ViT	93.26	93.30	79.82	83.58	89.87	87.41

TABLE IV
HIGHEST VALIDATION ACCURACIES IN 1000 TRAINING EPOCHS AND TESTING ACCURACIES AND THE EPOCHS WHEN THE HIGHEST VALIDATION IS REACHED

	MaxVal/%	Test/%			Epoch taken to reach the highest validation accuracy
		TOP–1	TOP–3	TOP–5	
FLS-ViR(p_fls=50)	72.53	70.37	90.37	94.07	286
FLS-ViR(p_fls=75)	77.07	77.78	93.33	94.07	192
FLS-ViR(p_fls=99)	75.47	75.56	94.07	97.78	161

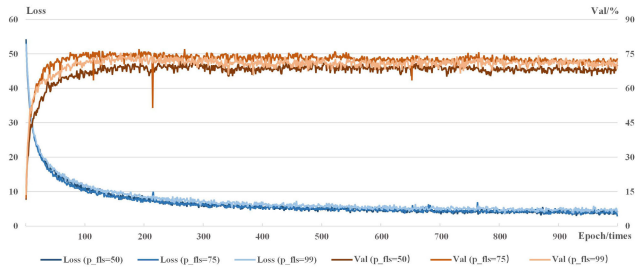


Fig. 5. Validation accuracies and losses in training.

that can be obtained by the model in the attention operation [see (6)]. For an input image of size 224 × 224 pixels, a patch size of 16 × 16 pixels is usually chosen, and the sequence length will be 196. Three length sizes of approximately 1/4, 1/3 and 1/2 of the input sequence were considered as untrained parameters for testing. The TOP-1, TOP-3 and TOP-5 were employed as the test accuracy. The validation and testing accuracies of three FLS-ViR models with different length sizes in 1000 training epochs were recorded (shown in Fig. 5). The epochs when the highest validation precisions occurred were also highlighted (shown in Table IV).

Parameter tests on fixed-length sequence length show that the best validation accuracy and TOP-1 testing accuracy can be obtained under the length of about 1/3 of the input sequence. With the precision of TOP-3 and TOP-5, around 1/2 of the sequence length obtained better results than the other two parameters. Especially under the TOP-5 precision, the classification result of length 99 surpasses the results of the other two experiments by more than 3%. Except for the TOP-5 accuracy which is the same as that with a sequence length of 75, a sequence length of 50 has the lowest accuracy among almost all tests. Based on the performance of the model on different sequence length parameters, the optimal solution for the sequence length parameter exists around 1/3 of the sequence length.

Moreover, we found that the sequence length is correlated with the speed of the optimal solution obtained by model training. As the sequence length increases, the highest verification accuracy will be obtained faster in model training. By collecting the losses and validation accuracies obtained in 1000 epochs, the convergence speed of the loss and validation accuracy is the fastest for a sequence length of 75. There is no significant difference in the speed of loss convergence for the three-parameter cases, however, the difference in accuracy convergence is more pronounced in the first 200 epochs. In summary, the convergence speed of precision and loss is related to the sequence length and is not associated with the count of epochs when the optimal weights are reached.

2) Size of the Input Image Effects: One of the main objectives of our model design is to reduce space complexity as well as decrease the memory on the graphics processing unit (GPU) for the same sequence length, which could make it possible to increase the input sequence length for the same GPU memory usage. In order to test the validity of the presented model, we selected three sizes of input images, 224 × 224, 448 × 448 and 512 × 512 with patch size of 16 × 16 and p_fls of 75 for evaluation and memory consumption testing. The size of the input image is labeled as p_siz in Table V. Based on the same well-trained weights, we employed the TJAID to train and test three different sizes of input images. Our model was deployed on a computer with an AMD Ryzen 9 16-core processor, a TUF-RTX3080-10G-GAMING display card and CUDA Version 11.2. By retrieving information from the NVIDIA system management interface, we were able to obtain long-term stable memory usage of the GPU during model training.

From the memory usage perspective, the model can effectively reduce memory usage, and the maximum image input of 600 × 600 pixels can be accepted with a 10G display card storage. It implies that, for the FLS-ViR model, a single display card can accept images from the TJAID with a size of 1024 × 1024 pixels with a sampling rate of less than 50%. When the number of pixels in the input image is raised by four times, i.e., p_siz is incremented from 112 × 112 to 224 × 224, the model has an increase in GPU memory usage by a factor of about 1.12 (from 2123 to 2371 MiB), while this figure is 1.56 (from 3477 to 5411 MiB) for the ViT model. The above experiments and analysis are based on the GPU performance, whereas based on the algorithm principle, since one of the inputs of the attention in FLS-ViR model is a

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TABLE V
HIGHEST VALIDATION ACCURACIES IN 1000 TRAINING EPOCHS AND TESTING ACCURACIES AND GPU MEMORY USAGE WITH DIFFERENT RESOLUTIONS

	MaxVal/%	Test/%			GPU memory usage/MiB
		TOP–1	TOP–3	TOP–5	
FLS-ViR(p_siz=224×224)	77.07	77.78	93.33	94.07	2371
FLS-ViR(p_siz=448×448)	76.80	74.81	91.11	97.04	5095
FLS-ViR(p_siz=512×512)	76.80	74.81	90.36	96.30	6237

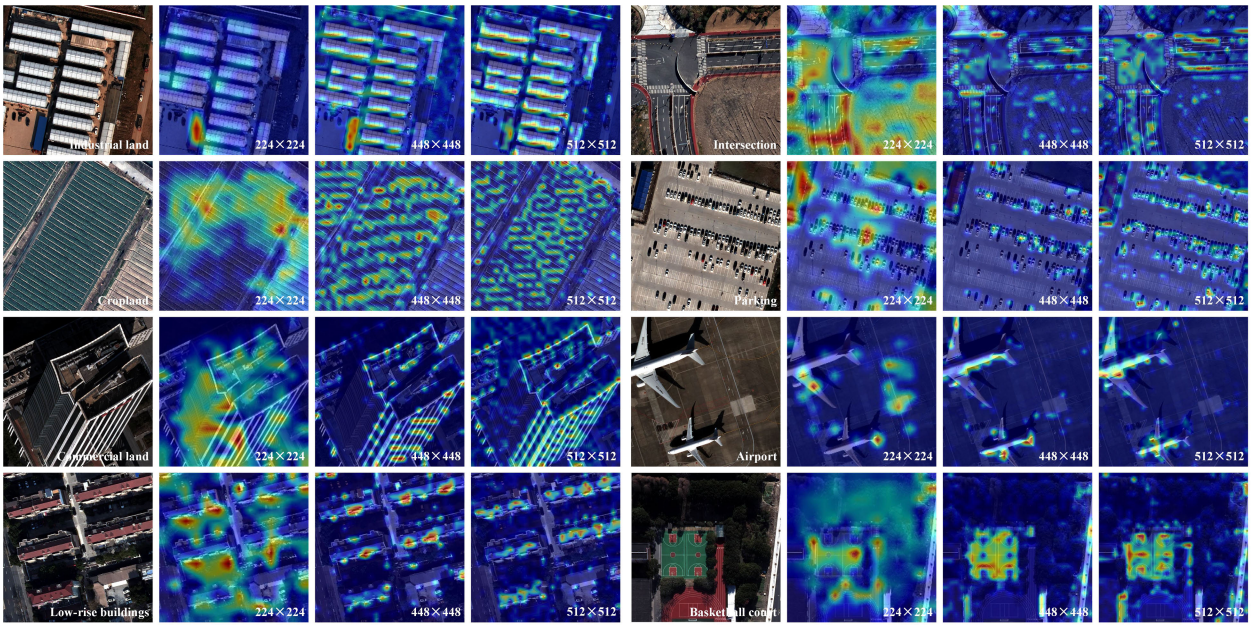


Fig. 6. Samples of heat maps for multiple categories with different input image sizes. The first and fifth columns are the original images. The second to fourth columns and the sixth to eighth columns are overlays of the heat map and the image corresponding to the input image sizes of 224 × 224, 448 × 448, and 512 × 512 pixels, respectively.

fixed-length sequence constructed under a constant parameter. Therefore, the space complexity of the proposed algorithm on the attention calculation is not affected by the growth of the sequence length, so the space complexity is $O(1)$.

From the accuracy improvement perspective, there is no significant accuracy change in the range of 224 × 224–512 × 512 pixels. Within 1000 epochs, the input training images of 448 × 448 and 512 × 512 pixels obtained the same maximum validation accuracy and TOP-1 test accuracy, while the input image of 224 × 224 pixels size was slightly higher than the other two in terms of maximum validation accuracy, TOP-1 and TOP-3 test accuracies. It is worth noting that the two training input images with larger sizes can obtain better results in TOP-5 test accuracy. We believe that for the task of scene classification, features are more significant on global features than local features, so the scaling of image size has less effect on the recognition results.

The positive impact of increasing the image size on feature recognition is further analyzed by means of heat maps. As can be seen from samples of some scene classes (see Fig. 6), the attention of the presented network shifts from continuous facets, to discrete points as the size increases. This is beneficial for features with small sizes in some scenes, such as cars in the parking lot. Under the condition of large image size images as training images, cars will be focused more precisely. For scene classes with significant texture features, such as cropland, or classes where we want the network to focus on identifying regularly aligned ground features, such as low-rise buildings, images with larger sizes can better help to capture texture information. In addition, heat maps show better attention to the necessary feature elements at larger input image sizes, such as faceted intersections and striped roads in the intersection category and the aircraft with entire form in the airport category. Especially in the cropland class, the small-size and large-size training images give rise to disparities in the model's attention to textures on planted and nonplanted land. Although the impact of different image sizes on recognition accuracy is not clearly demonstrated, we believe that these positive effects on the scene after expanding the input image size have the potential to enhance the improvement of the interpretation of details on remote sensing images.

3) Comparison With ViT and ViR: The performance of the proposed algorithm was measured in terms of accuracy and memory usage on two datasets compared with the other two algorithms. Pretraining weights for both ViT and ViR models were acquired on the ImageNet-21K dataset. The pretraining parameters for the FLS-ViR model were obtained in two steps. First, the weights are initialized by the Xavie method [59], the pretraining parameters obtained on the ImageNet-21K dataset are transferred to the decoder in the MAE model, and images were trained by the MAE method for 1000 epochs to generate weights. The encoder weights corresponding to

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TABLE VI
MAXIMUM VALIDATION ACCURACIES, TOP-1, TOP-3, AND TOP-5 TESTING ACCURACIES AND GPU MEMORY USAGE OF VIT, VIR, AND FLS-VIR ON TJAID AND AID

	TJAID				AID				GPU memory usage/MiB
	MaxVal/%	Test/%			MaxVal/%	Test/%			
		TOP–1	TOP–3	TOP–5		TOP–1	TOP–3	TOP–5	
ViT	89.07	87.41	94.81	96.30	93.46	98.00	99.30	99.50	5411
ViR	61.87	58.52	85.19	90.37	74.09	75.50	93.10	97.20	3077
FLS-ViR	77.07	77.78	93.33	94.07	78.66	80.60	94.40	97.40	2371

the best image reconstruction results will be selected as the pretraining weights in the next step. The training is then performed on the ImageNet_MINI dataset to classify images with the pretraining weights mentioned previously. The weight parameters corresponding to the highest verification accuracy will be used as the initial weights for the FLS-ViR model. Considering the training time and equipment factors, our training cannot be executed on a larger dataset, like the ImageNet-21K dataset with 14 million images and 21 000 classes. ImageNet_MINI is a smaller dataset, constructed by us using data from the ImageNet-21K dataset, which contains 38 400 images and 64 classes. The input image size is 224 × 224 pixels and the batch size is 16. Experimental results in Table VI demonstrate that our proposed method outperforms other algorithms in some aspects.

In the comparative analysis, we use ViR as a baseline model to show its improvement on ViT and our further enhancements. In terms of detection accuracy, our model improves the accuracy of ViR remarkably, particularly in maximum validation accuracy (upgrade 15.20%), TOP-1 (upgrade 19.26%), and TOP-3 (upgrade 8.14%) accuracy, on the TJAID (see Table VI). Meanwhile, the FLS-ViR model can close the scene classification with ViT in TOP-3 and TOP-5 accuracy. On the AID dataset, ViT still has the highest accuracies, and the FLS-ViR method has a better representation than ViR in both maximum verification accuracy and three test accuracies. Compared with the classification performance of TJAID, our model has a lead of more than 5% in TOP-1 accuracy than ViR. It is worth mentioning that the weights of our algorithm can obtain a validation accuracy of 46.91% on the AID in the first training epoch, based on the pretraining weight parameters, which are the training weights corresponding to the maximum validation accuracy acquired through the training on the TJAID. ViT can achieve fine classification results, due to its well-trained pretraining weights that we employed in experiments. In other words, robust pretraining parameters are one of the important factors for the model to attain great recognition results on the dataset. For the ViR model, the adaptability to image processing tasks may not be good enough. The FLS-ViR model provides a considerable improvement on the ViR algorithm in terms of scene classification performance, while it has also the potential to improve recognition accuracy based on more reliable parameters.

In terms of GPU memory consumption, the reduction in space complexity allows our algorithm to significantly reduce memory consumption. For either dataset, the model in this article has the same GPU memory consumptions with the same parameters, such as image size (224 × 224 pixels), patch size (16 × 16), and others. The reversible residual network realizes that the activations of layers in the encoder need not be stored in memory during backpropagation. According to the experimental results (see Table VI), the reduction in space complexity of ViR is noticeable versus ViT (dropped by 43.13%), thanks to the reversible residual network. The fixed-length sequence directly reduces the volume of input data for the encoder. Therefore, the proposed algorithm runs on GPU with the performance that, the memory usage of the FLS-ViR model is further reduced by 22.94% based on the optimization of ViR, which is 0.41 times that of ViT.

4) Performance of FLS-ViR on Different Categories: The recognition effect of the FLS-ViR model was tested on different scene categories in the TJAID. Based on the classification results of TOP-1 test accuracy, we built a confusion matrix map of 27 categories, as shown in Fig. 7. 13 categories achieved more than 90% recognition accuracy, nine categories gained more than 80% classification accuracy, and other categories attained more than 70% accuracy. Categories are easily misclassified into other categories that are similar to this category itself, e.g., low-rise buildings and high-rise buildings, playgrounds, and basketball courts. The best recognition performance is achieved in the boat category, and none of the images are misclassified. The worst classification is produced on the basketball court category with slightly over 70% accuracy. For some types that fall under agricultural land, the performance of the model is decent, with accuracies around 90%, which is attributed to the uniform global features. For the same reason, two types of forests, artificially planted woods and forests, can be well distinguished. For the categories with numerous buildings in the scene, the classification accuracy is independent of the regular arrangement. The most irregular class, the rural residential land, with the largest variety of features in the scene obtains the highest accuracy (98.04%) instead. The classification accuracy is lower for the categories with a linear distribution of ground features in the scene, such as road, railway, and bridge. Different recognition effects are indicated on some scene categories that contain sports fields. In the playground class, the drill ground itself occupies a larger portion of the image, while there are a variety of features in the basketball court class that lead to a small portion of sites.

The diversity of feature types impacts the recognition accuracy of the mode to some extent, but not absolutely; the distribution of ground features is not correlated with classification accuracy. Most of the scenes consisting of simple ground features achieved better recognition, however, some of these categories, such as the pond, obtained less than 80% discrimination accuracy. Meanwhile, many categories with

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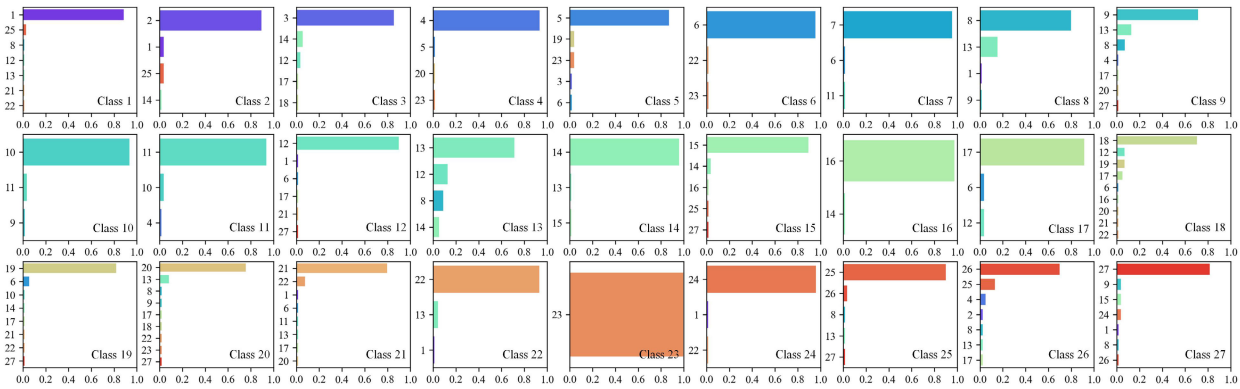


Fig. 7. Confusion matrix diagram of FLS-ViR on the TJAID. Each small graph indicates the percentage of categories classified into that category (the category number (seen in Table II) is in the bottom right corner, the horizontal axis expresses the percentage, and the vertical axis represents the class number of all categories classified into that category).

multiple features in the scenes, for example, the intersection and rural residential land, are precisely predicted. After the comprehensive analysis of the images against the heat map and the classification accuracy, the reason for the misidentification of the pond class is that the network focuses more on the roads at the edge of the pond than on the grid-like pools. In contrast, in the identification of rural residential land, the network catches the similarity of village houses in different images, hence its corresponding accuracy is superior. Through analyzing the relevance between the distribution of ground features and classification performance, the categories with accuracy over 90% and below 90% have almost the same number of scene categories with regular feature distribution. Therefore, the spatial arrangement of features does not affect the labeling of categories predicted by the FLS-ViR model.

Integrating the above analysis, it can be concluded that the recognition accuracy of scene types is determined by whether the model identifies the major elements finely. For example, the sandy ground in the golf course class can be better distinguished from the other unimportant features. Once the major ground features, as the iconic features of the scene categories, are accurately captured by the model, scene categories can be recognized with more than 90% accuracy.

5) Finetuning on Small Sample Datasets: We proved the applicability of our model on other remote sensing image datasets with a limited sample size, which will facilitate other scholars to employ the proposed model in their tasks. The fitness of our model on small sample datasets is tested on three similar datasets, RSC11, WHU-RS19, and UC MERCED (seen in Table I). The images in these datasets have high spatial resolution and the sample numbers are in the range of 1000–2000. A trainable adapter is added before the classifier, which is composed of linear operations and activation functions. The adapter and classifier are released for finetuning. The presented model converges easily within ten training epochs with the pretraining parameters we provided based on our dataset, as shown in Fig. 8. After 100 training epochs for these three datasets, RSC11, WHU-RS19, and UC MERCED achieved the highest validation accuracies of 80.98%, 90.75%, and 91.49%, respectively. There are differences in the classification results of our method for data with different distributions. The recognition of WHU-RS19 and UC

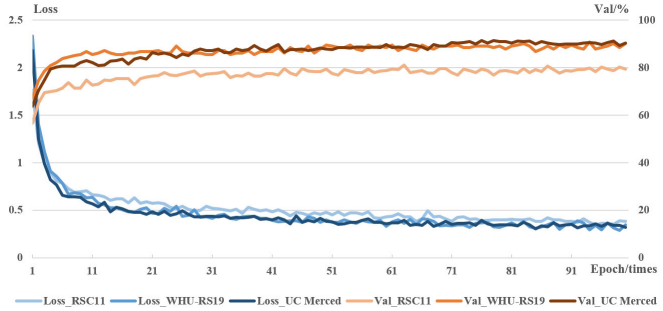


Fig. 8. Validation accuracies and losses in finetuning for three datasets, RSC11, WHU-RS19, and UC MERCED.

MERCED is close, while the classification results for RSC11 are significantly lower than the former two. This is partly due to differences in image quality and category separability.

V. Conclusion

With the mission of remote sensing image scene classification, this article establishes a high-resolution aerial image dataset containing 27 scene classes with a spatial resolution of 0.1 m and develops an FLS-ViR model. Compared to existing datasets, thanks to significant improvements in data richness, intraclass variation, and interclass similarity, TJAID is more challenging for algorithms. Compared with the Transformer-based image recognition methods, our proposed model has a remarkable reduction in space complexity and memory usage compared, based on the MSA encoder with fixed-length sequence and reversible residual structure. After introducing a self-supervised pretraining method, the proposed model can achieve a promising scene classification performance. We have demonstrated experimentally that it would enable the model to be trained with less GPU memory under the same input conditions as compared to other models, while allowing for large image sizes to be input, thus increasing the utilization of high-resolution remote sensing images. It was also proved through experiments that increasing the size of input images helps the attention-based model understand the key elements in the scene. Tests on different datasets also indicated that our model and the parameters trained on our dataset can be practiced on more high spatial resolution remote sensing image datasets with a limited sample size.

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We will further expand the dataset in both data volume and data type based on TJAID, according to different remote sensing image interpretation tasks. This will help us to apply the model to other downstream tasks, such as semantic segmentation. Meanwhile, our algorithm does not alleviate the problem of increased time complexity brought by the reversible residual network while space complexity is reduced. This makes our model take a longer time to train, which is what we will further improve. In addition, we focus more on the issue of spatial resolution in optical images in this article, and multispectral data input will also serve as our perspective for further exploration.

TJAID with blue, green, red, and near-infrared bands is available for download at: https://pan.baidu.com/s/1Tlinp4EPy8RGLSCFeL562Q. We also provide an open-source implementation of our method, FLS-ViR model, which is available at: https://github.com/LinHe1995/FLS-ViR.

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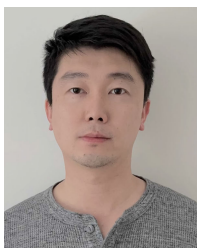
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